

PH.D. APPLIED QUALIFYING EXAM 2021 (TAKE HOME)

IMPORTANT - PLEASE READ THESE INSTRUCTIONS CAREFULLY BEFORE YOU START:

This is the Applied Part of the First-Year PhD Qualifying Exam in Statistics. **The exam consists of three problems - you are required to answer Problem 1 AND ONE OF Problems 2 and 3.** You are permitted to consult textbooks and other scholarly resources to help you in completing the assignment, but you **may not consult or discuss these problems in any way with anyone during the time that the exam is being administered.** If you draw on information from any outside resources (i.e., resources other than class material and notes) then you should clearly cite the source in your write-up.

You should complete the exam alone, in a room separate from any other test-takers. If you have technical questions about the two exam problems, you may email Prof. Zhiqiang Tan at ztan@stat.rutgers.edu (for problem 1) and Prof. Ying Hung at yhung@stat.rutgers.edu (for problems 2-3), cc'ing Prof. Harry Crane (hcrane@stat.rutgers.edu) AND Prof Tirthankar Dasgupta (td370@stat.rutgers.edu). If you have questions regarding logistics (e.g., submission instructions given below), please email Prof. Harry Crane (hcrane@stat.rutgers.edu) AND Prof Tirthankar Dasgupta (td370@stat.rutgers.edu).

Submission Instructions:

- (1) Type up your answers to the two problems in two separate Word or Latex files.
- (2) Create a folder for each problem. The folder must contain the following:
 - (a) A word or pdf file containing your answers to the problem, including all tables, figures and Appendices.
 - (b) An R (or python) file containing the codes used to generate your results.
 - (c) A word or a text file that gives a summary of the codes submitted.
- (3) Place the two folders into a single folder titled **yourlastname_yourfirstname_Appliedexam2021**. Zip this folder and email it to hcrane@stat.rutgers.edu AND td370@stat.rutgers.edu by 5 PM July 23, Friday.

Problem 1

Let (X_i, Y_i) , $i = 1, \dots, n$, be independent and identically distributed as (X, Y) , where Y is Bernoulli $(\frac{1}{2})$ taking value 0 or 1, and X is a 2×1 vector distributed as follows: X given $Y = 0$ is bivariate normal with mean vector $(-6, -6)$ and diagonal variance matrix $(5^2, 15^2)$, and X given $Y = 1$ is bivariate normal with mean vector $(6, 6)$ and diagonal variance matrix $(5^2, 10^2)$.

Suppose that the observed data are $S^\ell = \{(X_i, Y_i) : i = 1, \dots, n_1\}$ and $S^u = \{X_i : i = n_1 + 1, \dots, n\}$. That is, S^ℓ is a labeled dataset of size n_1 and S^u is an unlabelled dataset of size $n_2 = n - n_1$. Consider a logistic regression model:

$$(1) \quad P(Y = 1|X = x) = \frac{\exp(\beta_0 + \beta^T x)}{1 + \exp(\beta_0 + \beta^T x)},$$

where β_0 and β are unknown parameters. Let $(\hat{\beta}_0, \hat{\beta})$ be the maximum likelihood estimator (MLE) in model (1) based on only the labeled data S^ℓ , that is, a maximizer of the log-likelihood on the labeled data:

$$\sum_{i=1}^{n_1} \{Y_i \log \pi(X_i; \beta_0, \beta) + (1 - Y_i) \log(1 - \pi(X_i; \beta_0, \beta))\}.$$

Alternatively, define $(\tilde{\beta}_0, \tilde{\beta})$ as a solution to

$$0 = \frac{1}{n_1} \sum_{i=1}^{n_1} Y_i (1, X_i^T)^T - \frac{1}{n} \sum_{i=1}^n \pi(X_i; \beta_0, \beta) (1, X_i^T)^T,$$

where $\pi(x; \beta_0, \beta)$ denotes the right-hand side in (1). We are interested in studying the performance of $(\tilde{\beta}_0, \tilde{\beta})$ compared with $(\hat{\beta}_0, \hat{\beta})$.

For a simulation study, generate training and test data as follows in R (please run these codes exactly with the random seed):

```
#-----
set.seed(1)

### training data
n <- 1000
n1 <- 100

ys <- rbinom(n, size=1, prob=0.5)
ny0 <- sum(ys==0)
ny1 <- sum(ys==1)

xs <- matrix(NA, nrow=n, ncol=2)
xs[ys==0, ] <- cbind( rnorm(ny0, mean=-6, sd=5), rnorm(ny0, mean=-6, sd=15) )
xs[ys==1, ] <- cbind( rnorm(ny1, mean=6, sd=5), rnorm(ny1, mean=6, sd=10) )
```

```

# labeled data for training
cbind( xs[1:n1,], ys[1:n1] )

# unlabeled data for training
cbind( xs[-(1:n1),] )

### test data
ntest <- 10000

ytest <- rbinom(ntest, size=1, prob=0.5)
ntest.y0 <- sum(ytest==0)
ntest.y1 <- sum(ytest==1)

xtest <- matrix(NA, nrow=ntest, ncol=2)
xtest[ytest==0, ] <-
  cbind( rnorm(ntest.y0, mean=-6, sd=5), rnorm(ntest.y0, mean=-6, sd=15) )
xtest[ytest==1, ] <-
  cbind( rnorm(ntest.y1, mean=6, sd=5), rnorm(ntest.y1, mean=6, sd=10) )

# test data
cbind( xtest, ytest )
#-----

```

For all questions requiring coding, please provide both your R codes and output.

1) Explain why model (1) is misspecified in the preceding setting, that is, there are no values of (β_0, β) such that (1) holds for all x .

2) Report

- the estimates $(\hat{\beta}_0, \hat{\beta})$ on training data,
- the average log-likelihood and classification accuracy using the fitted model on test data.

The average log-likelihood on test data is the log-likelihood in model (1) divided by the test data size. The classification accuracy is the proportion of correctly classified observations, where x is classified as 1 if the fitted probability of $y = 1$ is greater than $\frac{1}{2}$, or classified as 0 otherwise.

3) Find a loss function $L(\beta_0, \beta)$ such that $(\tilde{\beta}_0, \tilde{\beta})$ can be equivalently obtained as a minimizer to $L(\beta_0, \beta)$.

4) Write your R function to compute the estimates $(\tilde{\beta}_0, \tilde{\beta})$, using Newton-Raphson.

5) Report

- the estimates $(\tilde{\beta}_0, \tilde{\beta})$ on training data,
- the average log-likelihood and classification accuracy using the fitted model on test data.

6) Plot the training data (labeled and unlabeled) and draw the following classification lines: the line based on $(\hat{\beta}_0, \hat{\beta})$, the line based $(\tilde{\beta}_0, \tilde{\beta})$, and the line based on the MLE in model (1) using **xs** and **ys**. Present your plot in an informative way as much as you can.

Problem 2

- (1) Assume that there are N observations from K classes in $p \geq 1$ dimensions. The p -dimensional input is in the form of an $N \times p$ matrix with each row denoting an observation and columns 1 to p representing p input variables. Let x_i denote the i th row of the input matrix. For each i , the class label is denoted by Y_i , where $Y_i \in \{1, \dots, K\}$ and $i = 1, \dots, N$. Show that the total covariance can be decomposed as follows:

$$(N-1)\Sigma_t = \sum_{i=1}^N (x_i - \hat{\mu})(x_i - \hat{\mu})^T = (N-K)\Sigma_w + (N-1)\Sigma_b,$$

where $\Sigma_w = \sum_{k=1}^K \sum_{Y_i=k} (x_i - \hat{\mu}_k)(x_i - \hat{\mu}_k)^T / (N-K)$ is the within class covariance matrix and $\Sigma_b = \sum_k N_k (\hat{\mu}_k - \hat{\mu})(\hat{\mu}_k - \hat{\mu})^T / (N-1)$ is the between class covariance matrix, N_k is the sample size in class k , $\hat{\mu}_k = \sum_{Y_i=k} x_i / N_k$ is the sample mean in class k , and $\hat{\mu} = \sum_{i=1}^N x_i / N$.

- (2) The idea of Fisher's reduced rank LDA is to find directions v by solving the following maximization problem:

$$\max \frac{v^T \Sigma_b v}{v^T \Sigma_w v}.$$

Show that this is equivalent to solving the following eigenvalue problem:

$$\Sigma_w^{-1} \Sigma_b v = \lambda v.$$

- (3) Randomly generate 100 observations from each of the following three classes in a 128-dimensional space using your own code. You will use the simulated observations in parts (4) and (5) of this problem.

Class 1: Gaussian with mean $\theta_1 = [3, 0, 0, \dots, 0]$,

$$P(X|Y=1) = N(\theta_1, \text{diag}(0.5, 2, 2, 2, \dots, 2)), \text{ and } P(Y=1) = 1/3.$$

Class 2: Gaussian with mean $\theta_2 = [0, -3, 0, \dots, 0]$,

$$P(X|Y=2) = N(\theta_2, \text{diag}(0.5, 2, 2, 2, \dots, 2)), \text{ and } P(Y=2) = 1/3.$$

Class 3: Gaussian with mean $\theta_3 = [-1, 0, 0, \dots, 0]$,

$$P(X|Y=3) = N(\theta_3, \text{diag}(0.5, 2, 2, 2, \dots, 2)), \text{ and } P(Y=3) = 1/3.$$

- (4) Based on the 300 observations in (3), write your own code to perform Fisher's reduced rank LDA. Create a visualization of your results by projecting the data onto the first two directions and label the three classes by different colors.
- (5) Perform PCA. Create a visualization of your results by projecting the data onto the first two principal directions and label the three classes by different colors. Comment on the difference between (4) and (5) and explain the reason in a couple of sentences.

Problem 3

The engine block and head joint sealing assembly is one of the most crucial and fundamental structural design in the automotive internal combustion engine. To best simulate the engine assembly process and operating conditions, a finite element model was developed to capture the complexity of part geometry, the compliance in the components, non-linear material material properties, and contact interface between the parts. Eight factors are selected for experimentation: gasket thickness (x_1), number of contour zones (x_2), zone-to-zone transition (x_3), bead profile (x_4), coining depth (x_5), deck face surface flatness (x_6), load/deflection variation (x_7), and head bolt force variation (x_8). A 27-run experiment is designed for these eight factors and the response is the gap lift (y).

The data obtained from this experiment can be found in the file

engine_experiment_data.txt

NOTE: Do NOT directly use the existing GP packages for the estimation and prediction while answering the following questions

- (1) Consider a Gaussian process model:

$$Y(\mathbf{x}) = \mu + Z(\mathbf{x}),$$

where μ is unknown, $\mathbf{x} = (x_1, \dots, x_8)^T$, $Z(\mathbf{x})$ is a Gaussian process with mean 0 and covariance function defined by $\text{cov}(Y((\mathbf{x} + \mathbf{h}), Y(\mathbf{x})) = \sigma^2 \phi(\mathbf{h}; \boldsymbol{\theta})$, and $\boldsymbol{\theta}$ is a vector of unknown correlation parameters. Define the correlation function $\phi(\mathbf{h}; \boldsymbol{\theta})$ by

$$\phi(\mathbf{h}; \boldsymbol{\theta}) = \exp\left(-\sum_{i=1}^8 \theta_i h_i^2\right).$$

Write down the mathematical expressions for MLEs for parameters μ , $\boldsymbol{\theta}$, and σ^2 (no proof required) and report the estimated values using the engine experiment data.

- (2) Write down the best linear unbiased predictor (BLUP) and the (plugin) predictive distribution (no proof required).
- (3) Based on the BLUP in (2), report the leave-one-out cross validation prediction error for the engine experiment data. Submit your codes following guidelines on the instructions page.
- (4) Change model (1) to

$$Y(\mathbf{x}) = \beta_0 + x_1\beta_1 + x_6\beta_2 + Z(\mathbf{x}),$$

where β_0 , β_1 , and β_2 are unknown coefficients. Repeat (2)-(3).

- (5) Suggest a final model that better fits the engine experiment data by comparing models (1) and (4) on the basis of suitable performance measures.

- (6) Based on the final model identified in (5), plot the BLUP as a function of x_1 and add the 95% predictive confidence bands. Submit the codes used to generate these bands.
- (7) Based on the final model, find the optimal settings of the eight factors that minimize the gap lift, explaining the algorithm and providing the code.